

Name:	ID:	Section:
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Question 1 [C03] [10 Points]

1	class A {
2	public int x, y = 2;
3	private double sum = 4;
4	public double getSum(){
5	return sum;
6	}
7	public double methodA(){
8	int temp = 3;
9	y = temp - x;
10	System.out.println(x+" "+y+" "+sum);
11	return sum + methodA(true);
12	}
13	public double methodA(boolean flag){
14	double y = -2;
15	if (flag == true){
16	this.sum = y + (++sum);
17	y = sum + 3 + this.y;
18	x += (int) sum * 2;
19	return y * 1.0;
20	}
21	return -1.5;
22	}
23	}
24	class B extends A {
25	public double sum = 3;
26	public static int temp = -4;
27	public B(){
28	int x = 2;
29	x = (int) (super.getSum() + (temp--));
30	sum = sum + methodA() + x + y;
31	}
32	public void methodA(int m, int n){
33	x = temp - 2 + m;
34	y = x - n;
35	sum = sum + super.y;
36	System.out.println(x+" "+y+" "+sum);
37	}
38	public String toString(){
39	return x+" "+y+" "+sum;
40	}
41	}

Illustrate the output of the following statements written in the main method of the driver code. [Answers without workings on the script will be rejected]

Driver code:

```
public class Exam4 {
    public static void main(String[] args){

        B b = new B();
        System.out.println(b);
        b.methodA(2,3);

    }
}
```

Output:

Name:	ID:	Section:
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Question 1 [C03] [10 Points]

1	class B {
2	public int x = 3, y;
3	private double sum = 4;
4	public double getSum(){
5	return sum;
6	}
7	public double methodB(){
8	int temp = 3;
9	y = temp - x;
10	System.out.println(x+" "+y+" "+sum);
11	return sum - y;
12	}
13	public double methodB(boolean flag){
14	double y = -2;
15	if (flag == false){
16	this.sum = y + (++sum);
17	y = sum + 2 + this.x;
18	x += (int) sum * 3;
19	return y * 2;
20	}
21	return -1.5;
22	}
23	}
24	class A extends B {
25	public double sum = 2;
26	public static int temp = -5;
27	public A(){
28	int x = 1;
29	x = (int) (super.getSum() + (--temp));
30	sum = sum + methodB(false) + x + y;
31	}
32	public void methodB(int m, int n){
33	x = temp + 1 - m;
34	sum = sum + methodB();
35	y = x + n;
36	System.out.println(x+" "+y+" "+sum);
37	}
38	public String toString(){
39	return x+" "+y+" "+sum;
40	}
41	}

Illustrate the output of the following statements written in the main method of the driver code. [Answers without workings on the script will be rejected]

Driver code:

```
public class Exam4 {
    public static void main(String[] args){

        A a = new A();
        System.out.println(a);
        a.methodB(4,1);

    }
}
```

Output:

Output for Set A

0	3	4.0
6	3	19.0
-5	-8	11.0

Output for Set B

12	0	16.0
-9	12	3.0
-9	-8	7.0

Rubric for Set A and Set B

SL	Criteria	Marks
1	Each correct output yields 1 mark	1 x 9 = 9
2	A trace table yields 1 mark	1
If no calculation/trace table is found, the student should receive 0 out of 10		